U.S. Middle East Strategic Withdrawal Response Model

Actor decisions, strategic pathways, and economic transmission | Version 1.2 | Live baseline: July 29, 2026

Decision question: If the United States materially reduces or ends its military and security posture in the Middle East, which actor responses become more plausible, and how might those responses transmit into energy, shipping, macroeconomic, and sovereign-risk stress?

Executive assessment

The simulation does not identify a single inevitable outcome. It finds a widening distribution of pathways as withdrawal removes residual capability and weakens security guarantees. Limited reductions are absorbed mainly through hedging and procurement. Full withdrawal and strategic disengagement increase the modeled frequency of competitive rearmament, maritime disruption, and regional escalation.

The most decision-relevant variables are not troop numbers alone. Withdrawal speed, residual intelligence and naval capability, and the credibility of remaining guarantees materially change actor incentives.

Model status: Live context / backtested / preliminary | Data quality: 96.4/100 | Live market stress: 63.6/100 (Elevated)

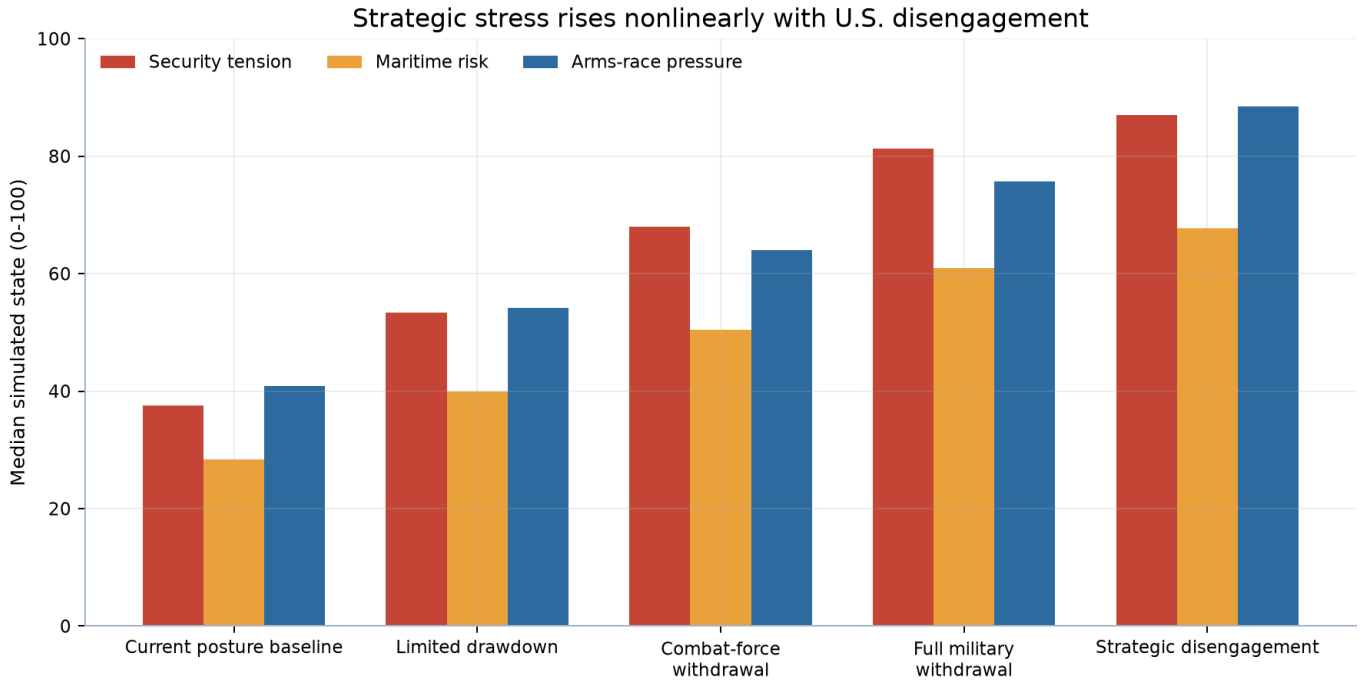


Figure 1. Median state values across Monte Carlo paths. The scale is an internally consistent stress index, not an observed probability.

Live economic baseline

The live context layer audits market freshness and macroeconomic coverage before presenting an observational baseline. It does not silently recalibrate the strategic simulation.

Current assessment: Elevated market stress (63.6/100) with Live data status and Backtested / preliminary calibration.

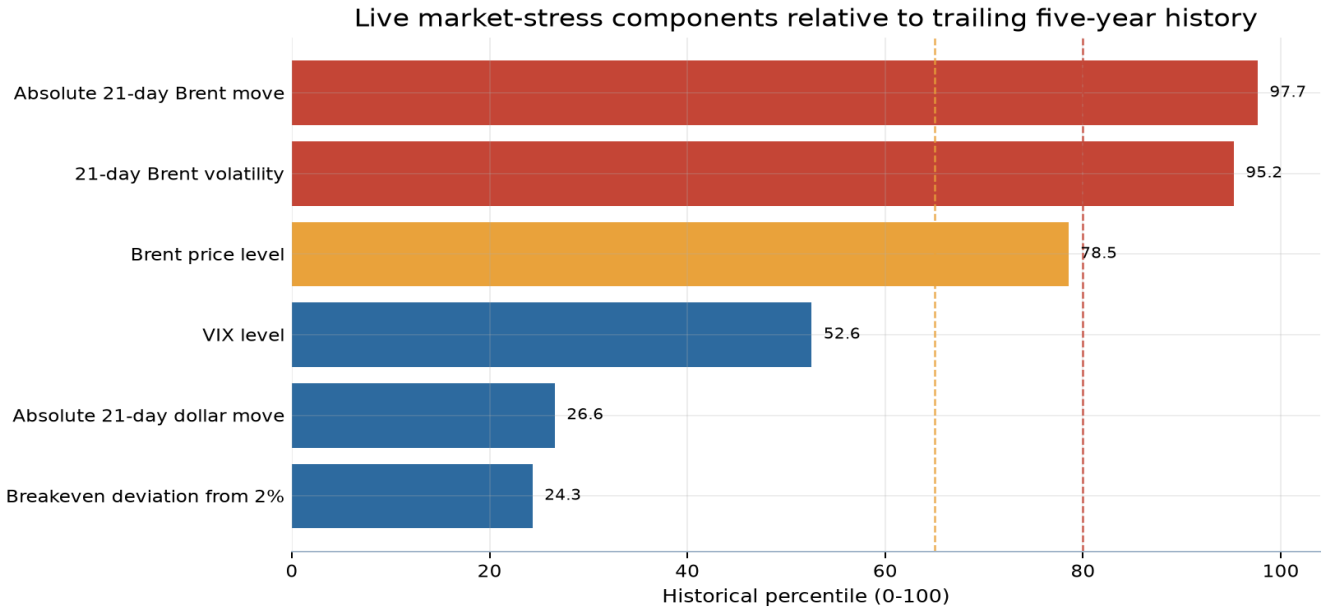


Figure 2. Percentiles compare current observations with their trailing five-year histories. They are stress indicators, not probabilities of military or political outcomes.

Live indicator	Latest value	Context
Brent crude	$91.82/bbl	21-day change +30.9%
Brent volatility	76.5%	21-day annualized
VIX	18.21	Equity-market volatility
10-year breakeven	2.26%	Market inflation expectation
Broad dollar index	120.71	21-day change -0.6%

Live Brent scenario overlay

Scenario	5th percentile	Median	95th percentile
Current posture baseline	$84.60	$91.68	$99.26
Limited drawdown	$91.48	$98.93	$106.24
Combat-force withdrawal	$98.05	$105.80	$113.37
Full military withdrawal	$104.63	$112.41	$119.92
Strategic disengagement	$108.68	$116.52	$124.13

The overlay applies simulated percentage changes to the latest observed Brent price. It is a conditional arithmetic translation, not a market-price forecast.

Scenario comparison

Scenario	WSI	Tension	Maritime	Arms race	Escalation	Brent median
Current posture baseline	0.00	0.38	0.28	0.41	0.0%	-0.1%
Limited drawdown	0.28	0.53	0.40	0.54	0.8%	+7.7%
Combat-force withdrawal	0.54	0.68	0.50	0.64	17.2%	+15.2%
Full military withdrawal	0.80	0.81	0.61	0.76	62.3%	+22.4%
Strategic disengagement	0.99	0.87	0.68	0.88	81.0%	+26.9%

WSI is the Withdrawal Shock Index. Escalation is the share of paths classified as regional escalation in the final simulation year. Brent changes are conditional stress ranges relative to the simulated baseline.

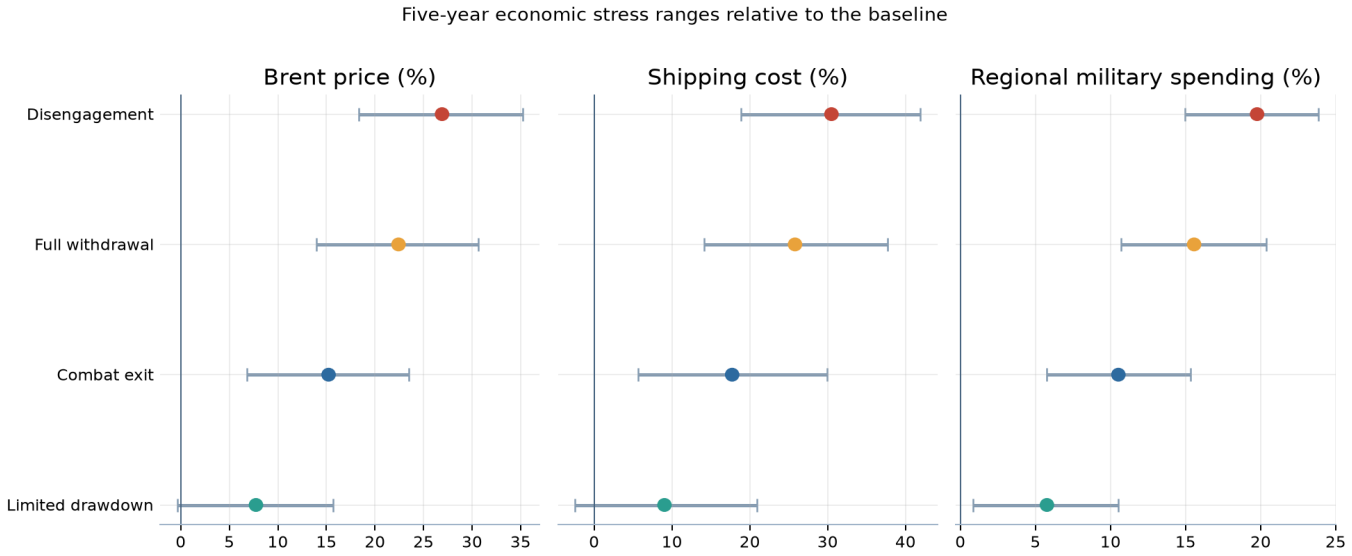


Figure 3. Dots show medians; whiskers show 5th-95th percentile ranges. Economic results use transparent structural stress coefficients and should not be read as causally identified forecasts.

Economic interpretation

The principal transmission chain is deterrence ambiguity to actor hedging or coercion, then to maritime risk and energy disruption, and finally to shipping costs, oil prices, inflation, growth, and sovereign risk. Oil exporters can gain revenue from higher prices while still incurring infrastructure, shipping, investment, and fiscal-volatility costs.

Historical calibration

Version 1.2 tests the Brent stress translation against sourced regional-security, energy-infrastructure, and maritime events. Event dates are sourced facts; the three shock-severity scores are versioned expert inputs.

Calibration status: Backtested / preliminary | 4/6 promotion gates passed | Historical coefficients remain diagnostic.

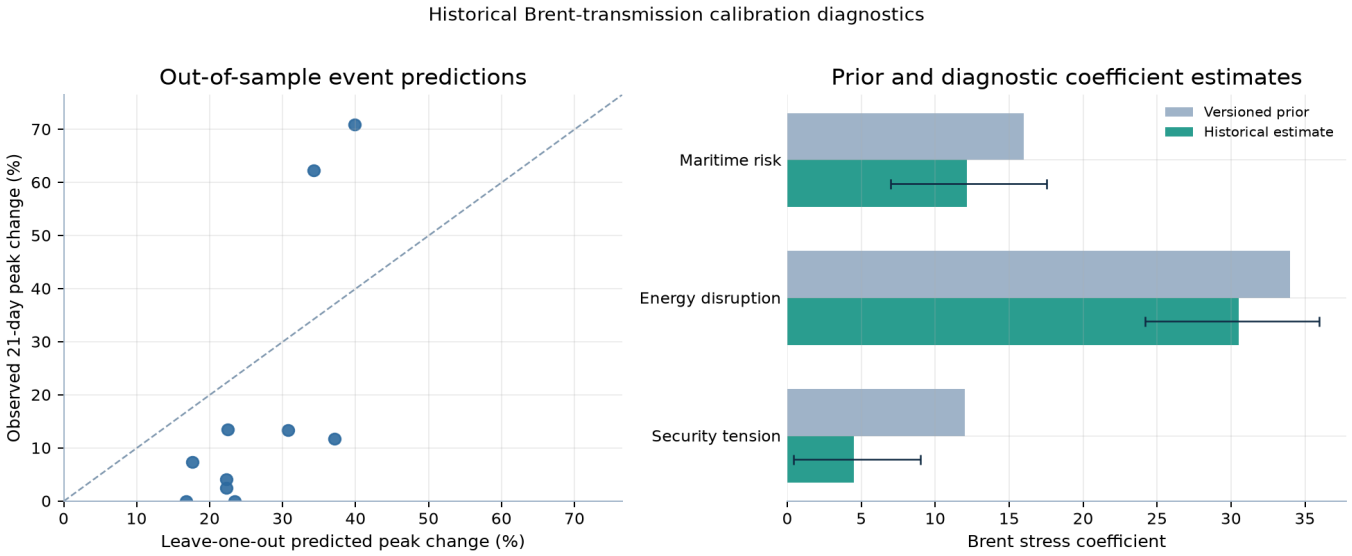


Figure 4. Left: observed peak Brent excursions versus leave-one-event-out predictions. Right: versioned priors versus historical estimates with 5th-95th percentile bootstrap intervals.

Promotion gate	Observed	Threshold	Result
Eligible events	10.00	>= 8.00	PASS
Source coverage (%)	100.00	>= 100.00	PASS
Brent coverage (%)	100.00	>= 80.00	PASS
Normalized LOO MAE	1.07	<= 0.75	HOLD
Directional agreement (%)	50.00	>= 60.00	HOLD
Maximum coefficient CV	0.58	<= 0.75	PASS

Leave-one-event-out MAE is 19.9 percentage points; directional agreement is 50%. The normalized-error and directional gates did not pass, so the model does not activate the fitted coefficients. This is a governed hold, not a failed run.

Conditional actor responses

Probabilities below are generated from actor characteristics, action requirements, and scenario conditions. They represent structured comparisons among configured choices, not measured intent.

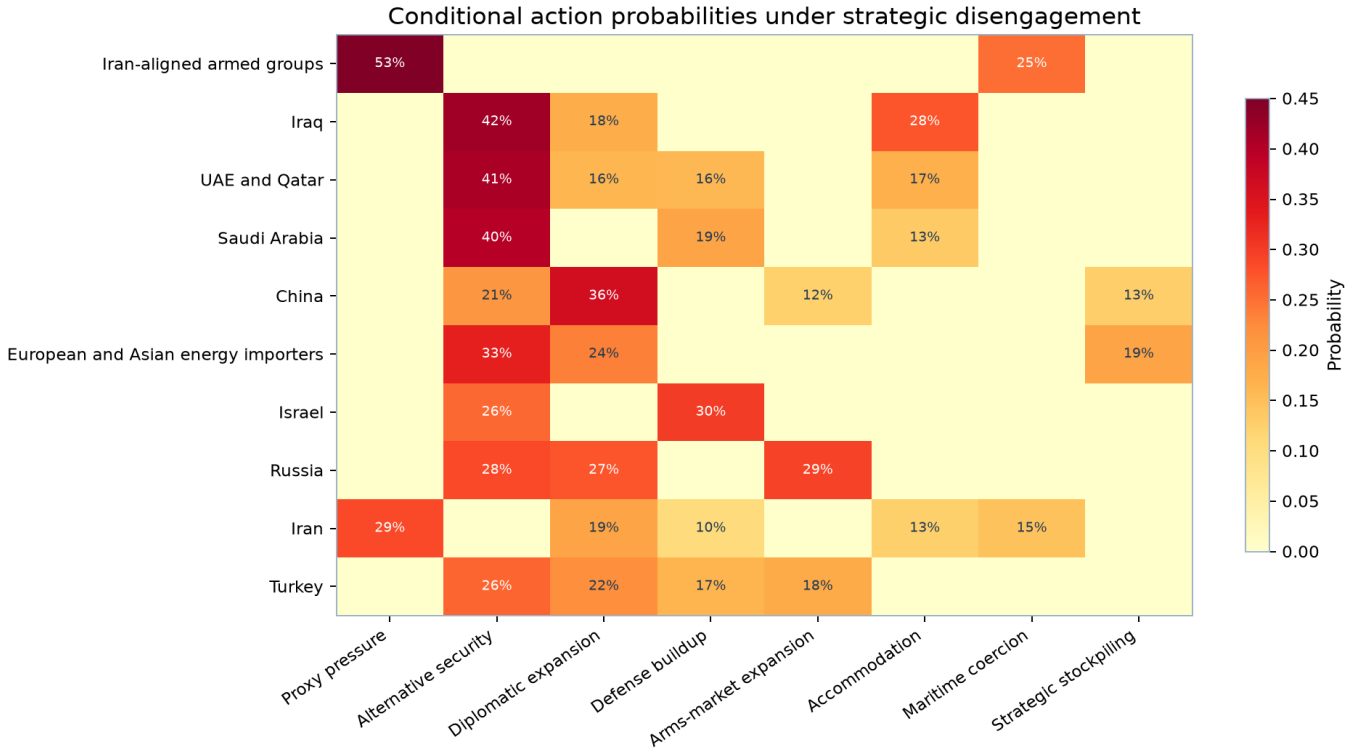


Figure 5. Higher values identify actions that become more competitive within an actor's configured choice set under strategic disengagement.

Actor	Highest-ranked configured action	Probability	Maintain
Iran-aligned armed groups	Expand proxy or partner pressure	52.6%	2.4%
Iraq	Seek alternative security partnerships	41.9%	1.6%
UAE and Qatar	Seek alternative security partnerships	41.1%	1.1%
Saudi Arabia	Seek alternative security partnerships	39.9%	0.9%
China	Expand diplomatic and economic influence	36.3%	1.9%
European and Asian energy importers	Seek alternative security partnerships	33.0%	2.3%
Israel	Increase defense spending	29.9%	0.8%
Russia	Expand arms-market position	29.3%	2.0%
Iran	Expand proxy or partner pressure	28.6%	1.4%
Turkey	Seek alternative security partnerships	26.1%	1.3%

Economic-security exposure

Vulnerability combines conflict proximity, U.S. security dependence, Hormuz exposure, energy dependence, fiscal capacity, reserve buffers, trade diversification, and the simulated strategic state. It measures sensitivity to the scenario, not the likelihood that the scenario occurs.

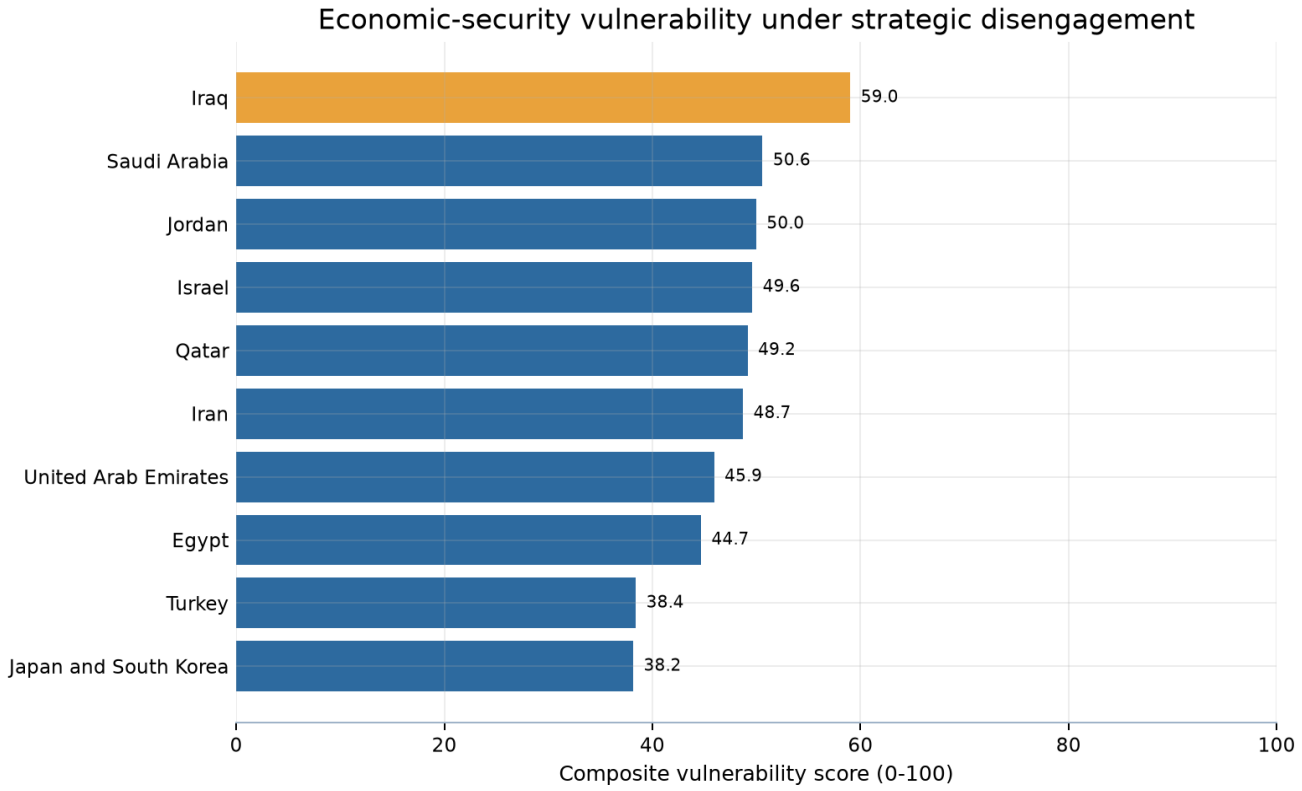


Figure 6. Highest composite exposure scores under the most severe configured scenario.

Key analytical cautions

Public-information constraint. Force posture, commitments, and private diplomacy are incompletely observed. **Behavioral uncertainty.** States and non-state groups are internally heterogeneous. **Calibration constraint.** Version 1.2 historical event windows are heterogeneous and did not pass every promotion gate. **Tail-risk constraint.** Nuclear use and other extreme outcomes are outside the modeled state space.

Method and sources

Withdrawal Shock Index: 35% force reduction, 20% withdrawal speed, 25% loss of residual capability, and 20% weakening of guarantees. Actor utilities are transformed with a softmax choice model. Each actor selects one action or maintains posture in every simulated quarter. Action effects update five persistent strategic states over a five-year horizon.

Reproducibility: All assumptions are stored in CSV files, random seeds are fixed by default, and the pipeline writes full probability, pathway, economic, vulnerability, and sensitivity tables.

Selected public baselines

Source	Use	Public URL
U.S. EIA	Hormuz flows, bypass capacity, and global oil outlook	eia.gov/world-oil-chokepoints
IMF PortWatch	Maritime trade and chokepoint disruption	portwatch.imf.org/pages/data-and-methodology
IMF WEO	Growth, inflation, fiscal, and external-sector context	data.imf.org/en/datasets/IMF.RES:WEO
World Bank	Country macroeconomic and governance context	data.worldbank.org
SIPRI	Military expenditure through 2025	sipri.org/databases/milex
FRED	Brent, volatility, inflation expectations, and dollar context	fred.stlouisfed.org

Full source register, assumption files, formulas, limitations, and validation roadmap are included in the project repository. Baseline verified July 29, 2026.

Prepared as an open-source strategic-economic research model. Findings should be reviewed by regional, military, energy, and macroeconomic specialists before any policy or investment use.